

Designing Enterprise Dataverse Data Models

Microsoft Dataverse & Power Apps

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Beginners

Power Apps

Citizen Devs

Analysts

Why Data Modeling Matters

What is Data Modeling?

A data model defines **what data** to store, **how it is structured**, and how tables relate to each other.

Think of it as the blueprint of your business application — before writing a single formula.

1

Foundation

Every Power App, Flow and report rests on the data model

2

Performance

Good models handle millions of rows without slowing down

3

Scalability

Add features without redesigning the whole database



Example: A Purchase Request app needs tables for Employees, Departments, Requests, Approvers, and Vendors — all connected.

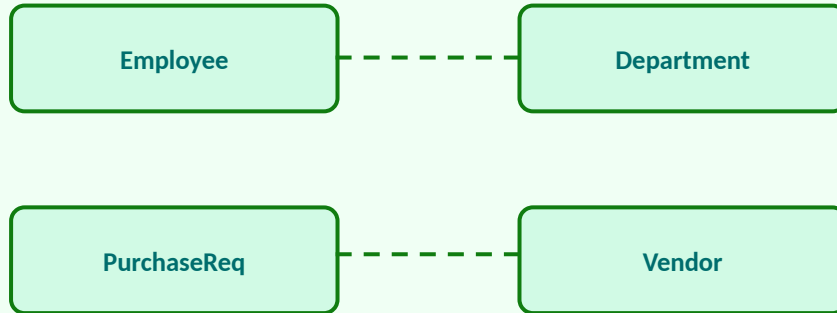
Multi-Table Data Modeling

✗ Single Table (Avoid)

EmpName	Dept	DeptHead
ReqTitle	Vendor	Price

- Data repeated on every row
- Hard to update — change in one row $\neq$ change everywhere
- Can't track relationships or history

✓ Multi-Table (Recommended)



- Each table stores one type of entity
- Update once, reflected everywhere
- Relationships managed by Dataverse lookups

Data Normalization Basics

Normalization removes duplicate data and organises it into related tables.

BEFORE (Duplicated Data)

Employee	Department	DeptLocation
Alice	Finance	Floor 3
Bob	Finance	Floor 3
Carol	Finance	Floor 3
Dave	IT	Floor 5

AFTER (Normalized)

Employee	Dept ID (FK)
Alice	D001
Bob	D001
Carol	D001
Dave	D002

Dept ID	Name	Location
D001	Finance	Floor 3
D002	IT	Floor 5

→ Normalize

✓ No repeated Department name or location

✓ Update Floor 3 → Floor 4 in one place only

✓ Lookup column links Employee to Department

Designing Relationships

One-to-One

Employee ↔ Employee Profile

Each employee has exactly one profile record

One-to-Many

Department ↔ Employees

One department contains many employees

Many-to-One

Purchase Requests ↔ Department

Many requests belong to one department

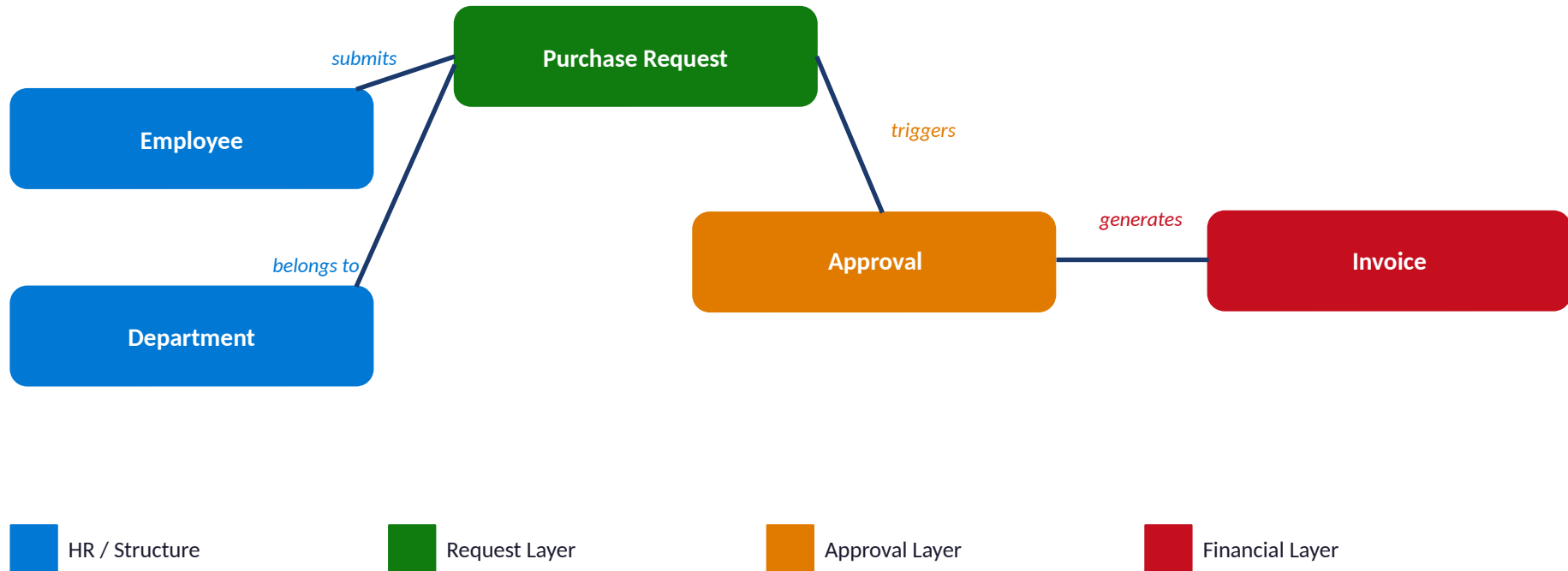
Many-to-Many

Requests ↔ Approvers

A request can have multiple approvers; an approver handles many requests

Business Workflow Relationships

How tables connect in a real enterprise workflow:



Validation Rules

Enforce data quality before records are saved to Dataverse.

Required Fields

Employee Name, Department — must not be blank

Number Validation

Amount > 0 and Amount ≤ 1,000,000

Email Validation

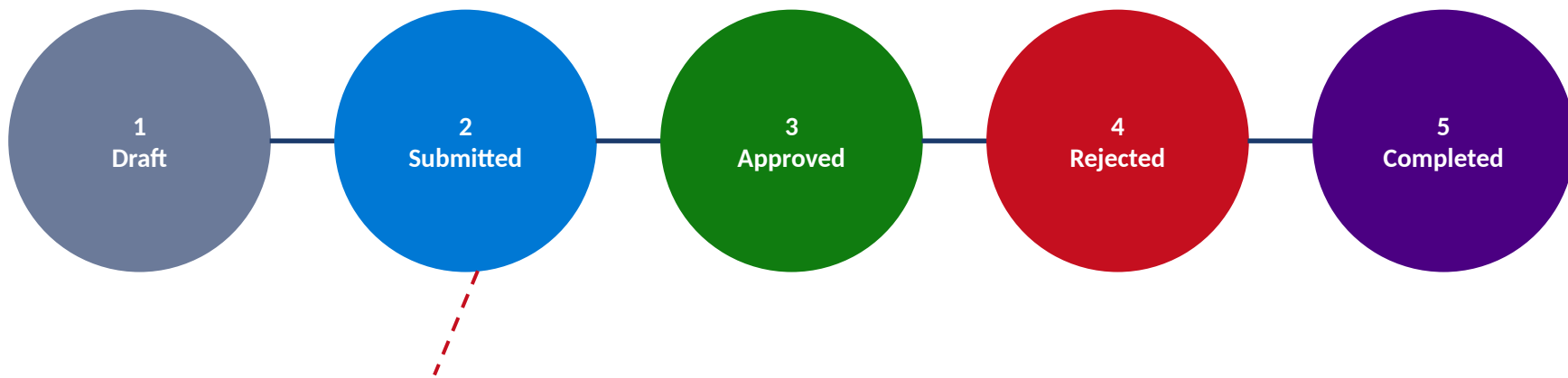
Pattern: `*@*.*` — e.g. john@company.com

Date Validation

Approval Date must be ≥ Request Date

Status Management — Record Lifecycle

Every business record moves through defined statuses. Dataverse Choice columns model this lifecycle.



↩ Rejected → back to Draft for revision

- ▶ Dataverse field type: Choice (formerly Option Set)
- ▶ Use Status Reason for sub-states (e.g. 'Pending Budget Check')
- ▶ Power Automate triggers run when status changes

Audit History Tables

Why businesses keep history instead of overwriting records



Compliance
Track every change for auditing



Traceability
See who changed what and when



Recovery
Restore previous values if needed

Sample: Purchase_Request_Audit Table

Audit ID	Request ID	Field	Old Value	New Value	Changed By	Changed On
AUD-001	REQ-100	Status	Draft	Submitted	Alice	2024-01-10 09:30
AUD-002	REQ-100	Status	Submitted	Approved	Manager Bob	2024-01-11 14:00
AUD-003	REQ-100	Amount	5000	7500	Alice	2024-01-12 10:15

Schema Changes and Their Impact

+ Adding Columns

Risk: Low

Power Apps automatically show new columns; existing records get the default value. Safe operation.

— Removing Columns

Risk: High

Breaks any Power App, Flow, or report that references the deleted column. Always check dependencies first.

✎ Renaming Fields

Risk: Medium

The logical name is permanent; Display Name can be changed safely. Logical name changes break integration.

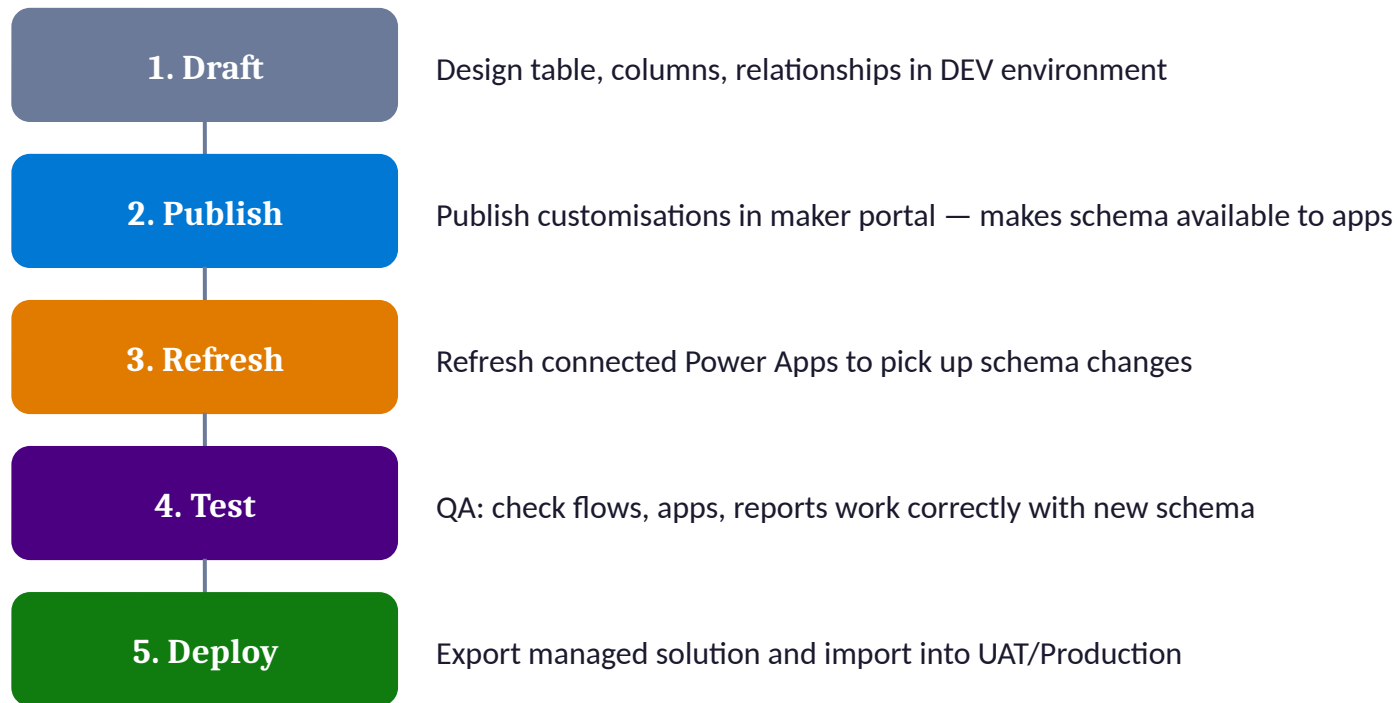
⚙ Changing Data Type

Risk: High

Not directly supported. Requires creating a new column, migrating data, and updating all dependents.

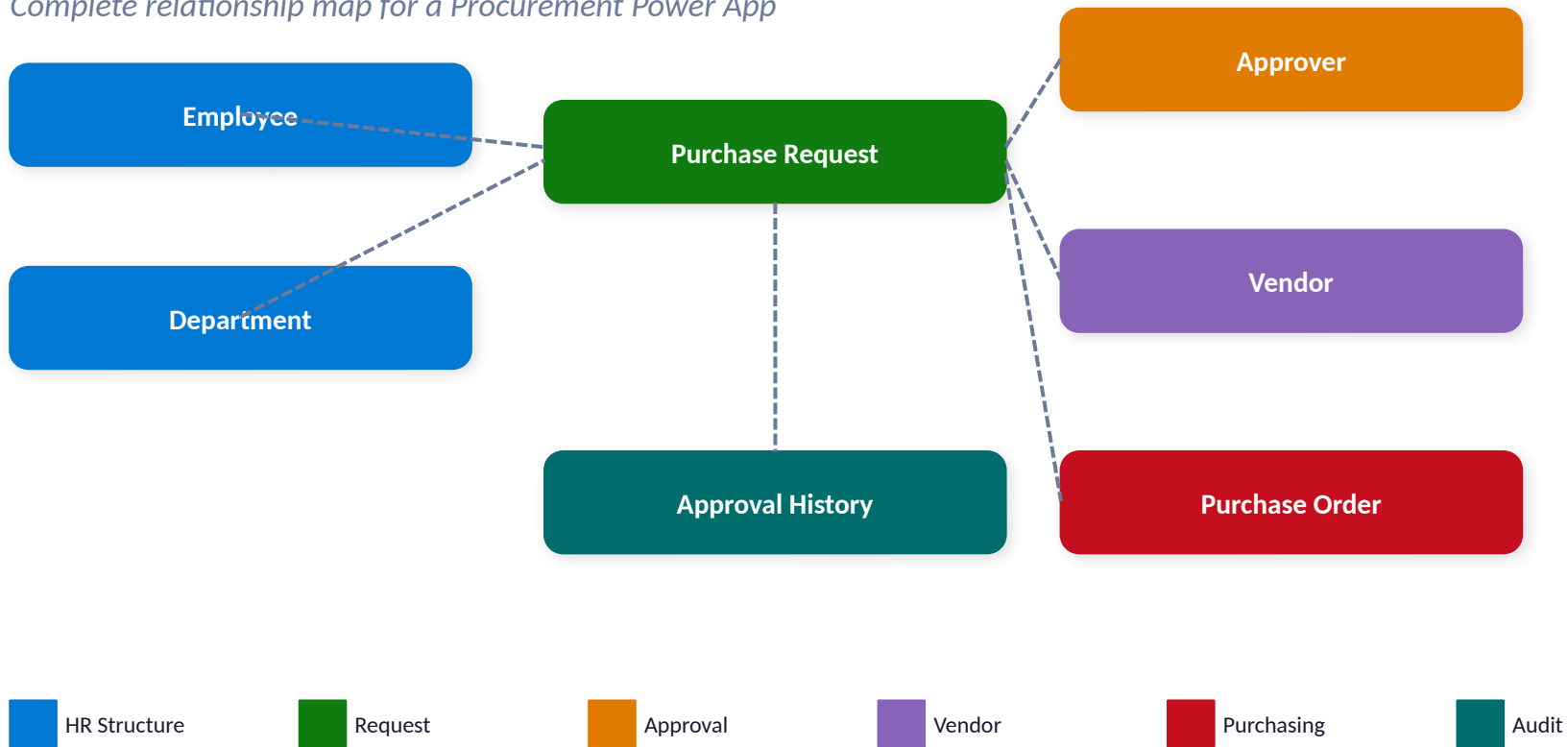
Schema Publishing — Deployment Flow

Follow this lifecycle when releasing schema changes to production.



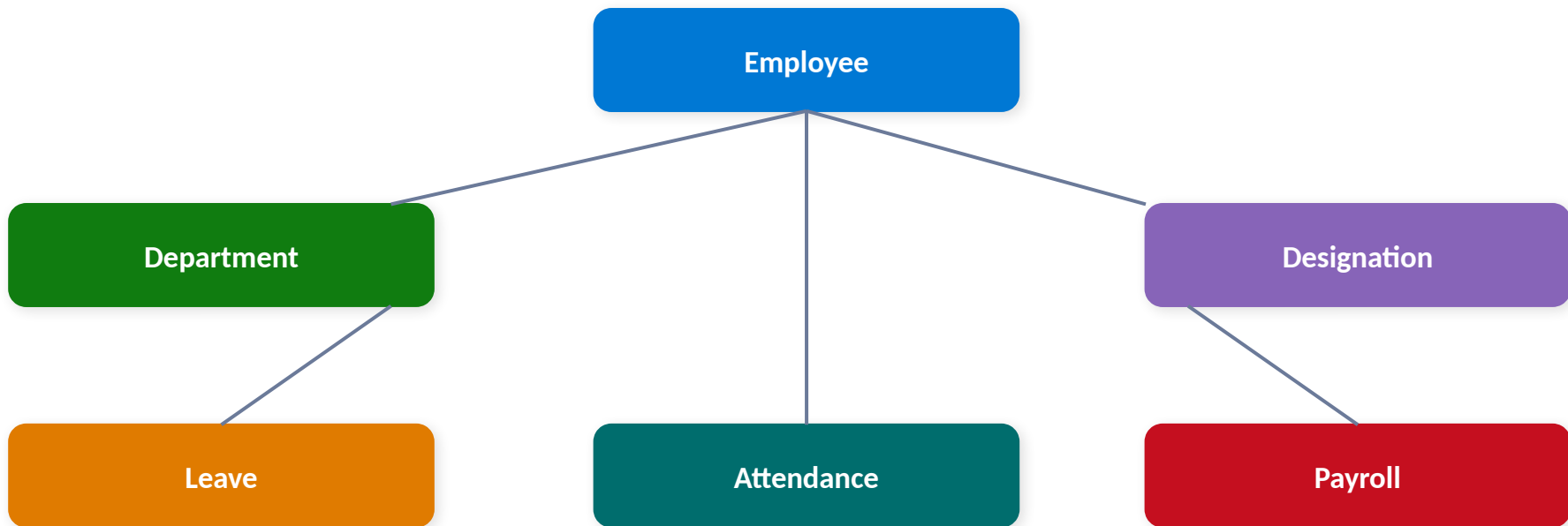
Enterprise Procurement Data Model

Complete relationship map for a Procurement Power App



Real-World HR Data Model

A complete HR system built on Dataverse tables



Employee: ID, Full Name, Email, Manager, Dept (Lookup), Designation (Lookup), Join Date

Leave: Employee (Lookup), Type, Start Date, End Date, Status, Approved By

Best Practices for Dataverse Data Modeling

Naming Conventions

Use PascalCase for table names, lowercase underscore for columns. Prefix custom tables with org acronym (e.g. contoso_Employee).

Lookup Columns

Always use Lookup columns to link tables — never store foreign keys as plain text. Lookups enforce referential integrity.

Choice Columns

Use Choice (Option Set) for status, type, category fields. Avoid free-text fields where values should be controlled.

Performance

Index columns used in filters and sorts. Avoid calculated columns on large tables. Use Alternate Keys for lookups.

Scalability

Design for growth from day one. Separate concerns into dedicated tables. Avoid wide tables (50+ columns).

Security Roles

Apply Column-level security for sensitive fields (Salary, NID). Use Owner or Team ownership model per table.

Common Design Mistakes to Avoid

✗ Common Mistake

✗ Storing everything in one giant table

✗ Using text fields for status values

✗ Deleting columns that apps still reference

✗ No naming conventions

✗ Skipping validation rules

✗ Not planning for audit/history

✓ Best Practice Fix

✓ Separate entities into dedicated tables

✓ Use Choice columns for controlled vocabularies

✓ Check Solution Dependencies before deleting

✓ Define and enforce naming standards at project start

✓ Add Business Rules for required fields and formats

✓ Build Audit tables from day one

Business Case Study — From Chaos to Dataverse

Contoso Ltd. had a procurement process run entirely on Excel. Here is how they migrated.

BEFORE: Excel-Based

- ✗ Single Excel file shared via email
- ✗ No approval workflow — email chains
- ✗ Duplicate vendor entries in every row
- ✗ No audit trail — cells overwritten
- ✗ Reports: manual copy-paste monthly
- ✗ Version control: none



AFTER: Dataverse Solution

- ✓ Separate tables: Employee, Dept, Request, Vendor
- ✓ Power Automate approval workflow
- ✓ Vendor master table — single source of truth
- ✓ Audit History table — every change logged
- ✓ Power BI dashboard — real-time reporting
- ✓ Version control via ALM Solutions

 Result: 70% reduction in approval cycle time | 100% audit compliance | Zero duplicate records

Hands-on Activity — Extend the Procurement App

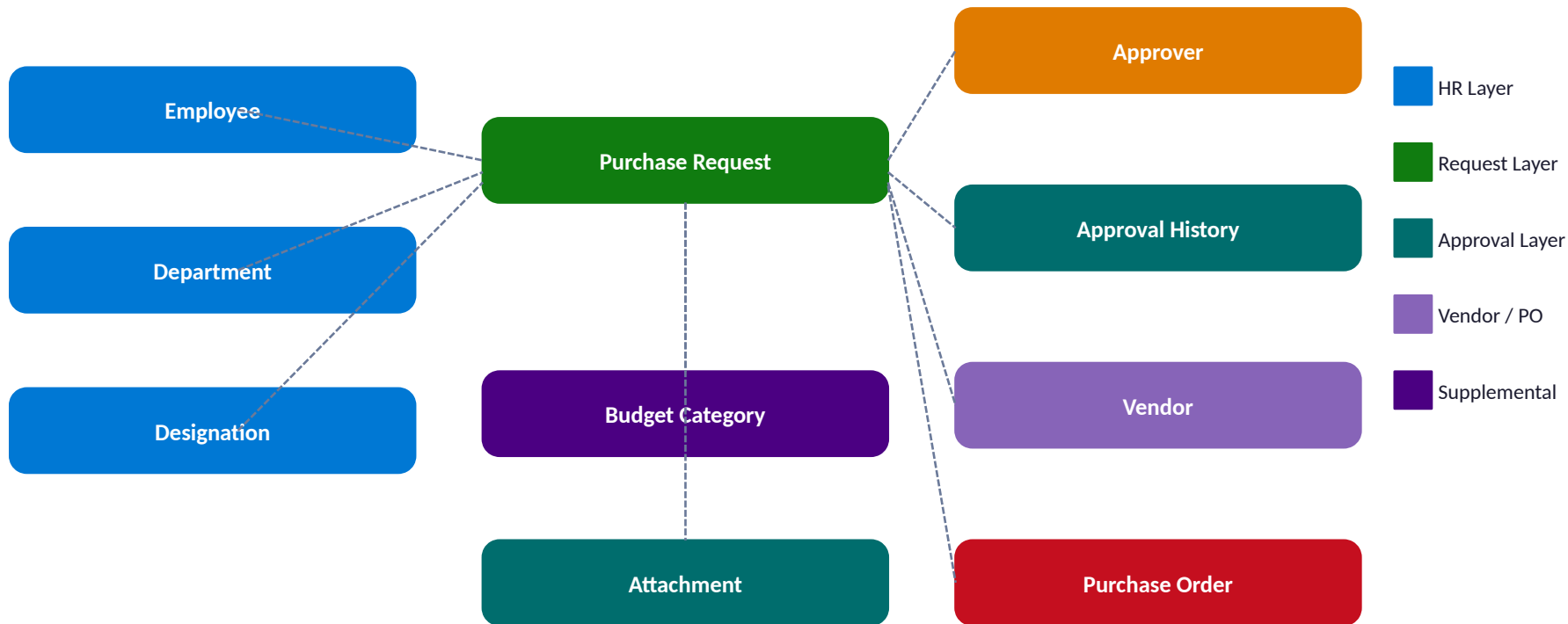
Add the following tables and columns to the procurement data model you built:

#	Field/Table	Type	Values / Notes	Why Needed
1	Approval Level	Choice	L1, L2, L3, Board	Route request to the right approver tier
2	Budget Category	Choice	CAPEX, OPEX, Emergency	Classify spend type for reporting
3	Cost Center	Lookup → Dept	Links to Department	Enables departmental budget tracking
4	Comments	Multiline Text	Max 2000 chars	Free-text note from approver or requestor
5	Approval History	Related Table	1:N from Purchase Request	Audit trail of each approval action
6	Attachment Metadata	Table	FileName, URL, UploadedBy, Date	Track supporting documents



Open make.powerapps.com → Select your environment → Tables → Add the fields above to your Procurement solution

Complete Dataverse Architecture — All Tables Connected



Summary — Key Concepts Learned

01

Data Modeling Fundamentals

Structure data into tables; separate entities

02

Normalization

Eliminate duplication; one source of truth

03

Relationships

1:1, 1:N, N:N via Lookup columns

04

Validation & Status

Business Rules, Choice fields, lifecycle management

05

Schema Management

Publish, test, deploy via ALM Solutions

06

Real-World Models

Procurement & HR reference architectures

07

Best Practices

Naming, performance, security, scalability

08

Hands-on Application

Extended procurement model with audit & metadata

Thank You!

Questions & Answers

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